CS-Exebit 2025 Quantathon

May 1, 2025

Instructions

- You are given 1 hour to solve the problems.
- There are 15 problems in total.
- If a probability is given in the form $\frac{x}{y}$, it means that greatest common divisor of x and y is 1.

Enjoy Solving!!!

1 Rolling Dice and Divisibility (1 point)

Two fair six-sided dice are rolled. The probability that the product of the two numbers rolled is divisible by 6 is $\frac{x}{y}$, what is x + y?

2 Choosing Balls with Restrictions (2 points)

An urn contains 5 red balls, 4 blue balls, and 3 green balls. Two balls are drawn randomly without replacement. The probability that the two balls are of different colors is is $\frac{x}{y}$, what is x + y?

3 Karthikeya and Yash's Paper Problem (4 points)

Karthikeya and Yash are writing questions for a Math Competition at a rate of 1 per minute. Immediately after Karthikeya writes a question, Yash eats his paper with probability $\frac{1}{7}$, so that Karthikeya must restart.

After an extremely long time (assume infinite), Datha walks in. What is the expected number of questions Datha sees written on Karthikeya's paper?

4 Exebit Lottery (2 points)

In the Exebit Lottery, 50 balls are drawn at random from $n > 100$ balls numbered 1 through n. If the probability that no pair of consecutive numbers is drawn is equal to the probability that exactly one pair is drawn, find n.

5 Can you Solve? (4 points)

Let $a \diamond b = ab - 4(a + b) + 20$. Evaluate

$$1 \diamond (2 \diamond (3 \diamond (\cdots (2020 \diamond 2021) \cdots))).$$

6 Reversing Digits and Doubling (2 Points)

A two-digit positive integer, n, is doubled and then added to 2 to obtain N. Then, the digits of n are reversed to form a new positive integer m. Given that $N = m$, find n.

7 Smallest Sum of Distances to Two Points (3 points)

Let ℓ be a line and let points A, B, C lie on ℓ so that $AB = 7$ and $BC = 5$. Let m be the line through A perpendicular to ℓ. Let P lie on m. Compute the smallest possible value of $PB + PC$.

8 Is it easy? (4 points)

Let $a_0 = -2$, $b_0 = 1$, and for $n \geq 0$, let

$$a_{n+1} = a_n + b_n + \sqrt{a_n^2 + b_n^2}$$

$$b_{n+1} = a_n + b_n - \sqrt{a_n^2 + b_n^2}$$

If a_{2020} can be represented in the form $a^x\sqrt{b} - c^y$, where a, b, c are integers such that a, c are minimized and c is square free, find $x + y$.

9 Observe!!! (4 points)

Distinct prime numbers p, q, r satisfy the equation

$$2pqr + 50pq = 7pqr + 55pr = 8pqr + 12qr = A$$

for some positive integer A. What is A?

10 Minimum Roll (3 points)

Imagine you roll three fair 100-sided die. What is the expected value of lowest roll?

11 Five Consecutive Heads (4 points)

A coin is tossed until Five Consecutive Heads appear. what is the expected number of trails to achieve this?

12 Dice Game (4 points)

There are 6 players, numbered 1 to 6. Player 1 rolls a die. If player 1 rolls a 1, he wins and the game ends. If Player 1 rolls any other number, the die is passed to player corresponding to the rolled number, and the game continues with that player rolling. The game end when the player rolls their own number.
The probability that player 1 wins the game is $\frac{x}{y}$, what is x + y?

13 Deck of Cards (2 points)

A deck of cards has 100 playing cards, 10 cards of each number from 1 to 10. This deck is well-shuffled. You keep turning up cards until you see the number 1 for the first time. On average, how many cards are you expected to turn?

14 Jim and Die (4 points)

Jim will roll a fair, six-sided die until he gets a 4. What is the expected value of the highest number that he rolls through this process?

15 Two Player Game (3 points)

Two players take turns rolling two six-sided dice. Player A goes first, followed by player B. If player A rolls a sum of 6, they win. If player B rolls a sum of 7, they win. If neither player rolls their desired value, the game continues until someone wins. The probability that Player A wins is $\frac{x}{y}$, what is x + y?